

Supplementary Material

Scale-Explicit and Uncertainty-Qualified Assessment of Tangent-Only Matrix Perturbations in Coupled Finite-Element Linearizations

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Purpose and scope

This supplement preserves exact numerical values, uncertainty components, precision and identity checks, tooling provenance, and the complete claim–evidence matrix supporting the main manuscript. M8 and M16 are independent frozen finite-element linearizations rather than a mesh or refinement sequence. The material below documents the existing calculations; no new scientific case was executed for the public-release v0.6 revision.

1 Synthetic analytical verification

Table S1: Pre-specified reproduction of the P2 directional construction.

ε	μ_J (matched)	ρ_1 (matched)	ρ_g (matched)	aligned η_z	orthogonal η_z
1e-12	1.0000e-12	1.0000e-12	1.0000e-12	1.0000e-12	0
1e-10	1.0000e-10	1.0000e-10	1.0000e-10	1.0000e-10	0
1e-8	1.0000e-08	1.0000e-08	1.0000e-08	1.0000e-08	0
1e-6	1.0000e-06	1.0000e-06	1.0000e-06	1.0000e-06	0
1e-4	1.0000e-04	1.0000e-04	9.9990e-05	9.9990e-05	0
1e-2	1.0000e-02	1.0000e-02	9.9010e-03	9.9010e-03	0

Table S2: Pre-specified P3 coordinate-transformation and redeclaration calculation.

s	transformed $\widehat{\Delta J}_{22}$	transformed physical $(\Delta J)_{22}$	transformed η_z	redeclared physical $(\Delta J)_{22}$	redeclared η_z
1	1.0000e-06	1.0000e-06	1.0000e-06	1.0000e-06	1.0000e-06
10	1.0000e-05	1.0000e-06	1.0000e-06	1.0000e-07	1.0000e-07
1e3	1.0000e-03	1.0000e-06	1.0000e-06	1.0000e-09	1.0000e-09
1e6	1.0000e+00	1.0000e-06	1.0000e-06	1.0000e-12	1.0000e-12

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2 Frozen operand provenance

Case	Dimension	Snapshot SHA-256	Declared operand contract
M8	196	ACF00D823AAE 8BC073F14ECF A322AED1B004 CBBA85D5FA83 416FEC4D24D87482	Frozen IEEE binary64 J , J_g , b , free map, and scales; transformed free coordinates; Euclidean norm.
M16	900	22A51CA5BB44 03D1EB9499F4 588D6EF19CE7 D6C3636429D5 EE2FCA6AE3AB207E	Independently frozen IEEE binary64 J , J_g , b , free map, and scales; transformed free coordinates; Euclidean norm.

The M8 and M16 matrix snapshots were each reproduced bitwise by an independent capture witness. Each offline application reconstructed every declared binary64 operand exactly before high-precision comparison. The perturbation identity $J_g = J + \Delta J$ was checked elementwise before either response route was classified.

3 Exact route-specific results

Table S3: Exact M8 and M16 route-specific values copied from the registered decision files.

Case	Route	Quantity	Exact registered value
M8	Direct difference	Observed response	0.0000000000029185014536 758156190032956007920590 207248424420081982240251 520232804179861608795259 82636481996805201
M8	Direct difference	Uncertainty bound	0.0000000000005880137782 404462163207715781856199 927166001714634199187384 669088463464446774100523 218868676802748951
M8	Direct difference	Lower interval bound	0.00000000000023304876754 353694026825240226064390 280082422705447783052866 851144340715414834694736 60749614316530306
M8	Direct difference	Upper interval bound	0.00000000000035065152319 162618353240671789776790 134414426134716181427636 189321267644308382895783 04523349677080096
M8	Direct difference	Guaranteed relative-error upper bound	0.2523136184921452489334 362403820407858086971785 578796069759756449422942 232194245526536411060357365522
M8	Direct difference	Decision	DISTINGUISHABLE_FROM_ZERO
M8	Response equation	Observed response	0.0000000000029200142254 489211942473568021747142 861853194538287404086512 718756577010642403187437 02369440071251828

Case	Route	Quantity			Exact registered value
M16	Response equation	Lower interval bound			0.0000000000246367184540 401403630682614743809968 333853654899273031974363 982197781281200726780876 353465113766222
M16	Response equation	Upper interval bound			0.0000000000246367184679 011055974029738346233036 657181318216038385307190 121315169653103418235836 469238192954816
M16	Response equation	Guaranteed bound	relative-error	upper	0.0000000002813070511032 623424952586212313222893 810817729228669242584304 341140586873408936987001 183002654740493
M16	Response equation	Decision			DISTINGUISHABLE_FROM_ZERO

4 Precision, residual, identity, and repeatability evidence

Table S4: Independent-reference, identity, and repeatability gates.

Case	Audit	Object	Maximum registered difference	Gate	Status
M8	Precision ladder	120/180 dps	8.8057092380 884434187176 962461947943 092333470546 718574519151 728781821208 054678586278 82320352049155351e-121	$\leq 10^{-80}$	PASS
M8	Reference residual	Parent and response systems	2.5509756331 440603586511 214959945542 624542125528 335796688014 351623728996 256868809424 33077965138698482e-124	$\leq 10^{-90}$	PASS
M8	Response identity	Parent difference versus response solve	9.1849085476 145843658430 984457176152 739279490697 844155754981 011387453725 230772676985 99848757236500405e-123	$\leq 10^{-80}$	PASS
M8	Repeatability	120/180 dps independent process	0; 0	$\leq 10^{-80}$	PASS
M16	Precision ladder	120/180 dps	3.9700701327 504386317003 831479435048 947013122943 519143716924 264139187506 177819881847 65817763642002851e-121	$\leq 10^{-80}$	PASS

Case	Audit	Object	Maximum registered difference	Gate	Status
M16	Reference residual	Parent and response systems	4.1440816367 892560429801 527838154080 514775758958 038539094508 388888254696 147598977293 45099133667671501e-124	$\leq 10^{-90}$	PASS
M16	Response identity	Parent difference versus re- sponse solve	3.0422633711 275487636633 980676903715 062449817277 763274882660 901630223123 725003858668 54677112813264999e-122	$\leq 10^{-80}$	PASS
M16	Repeatability	120/180 dps independent process	0; 0	$\leq 10^{-80}$	PASS

The reference precision ladder used 120 and 180 decimal digits for each application. M8 consumed exactly 467.59207850007806 s with recorded peak memory of 115154944 bytes. M16 consumed exactly 37512.160941799986 s with recorded peak memory of 1452195840 bytes. These values document two of-line applications; they are not a complexity fit or an online-estimator benchmark.

5 Route-specific uncertainty components

Table S5: Route-specific uncertainty-component coverage.

Case	Route	Component	Registered component bound	Status
M8	Direct difference	Parent-z	0.0000000000003226016562 676377291832930729419859 852938628308420601477443 639054955629477059647754 299784885800860621	COVERED
M8	Direct difference	Parent-z _g	0.0000000000002654033664 645770837841684095747019 844356589723578280133643 990225258788469947115176 298525210742069229	COVERED
M8	Direct difference	Final subtraction	0.0000000000000000087555 082314033533100956689320 229870783682635317576297 039808249046499767337592 62055858025981910067742	COVERED
M8	Response equation	Forcing formation	0.0	COVERED
M8	Response equation	Inherited parent	0.00000000000000000000 000011461113170257339148 722517767779506858043093 261374428512921636009578 312627822241702445839755 60054732	COVERED

Table S6: Expanded v0.4 claim–evidence matrix. The CSV is authoritative.

ID	Tier	Claim	Allowed strength	Prohibited extrapolation
C01	T1_THEOREM	Matrix-space smallness alone is insufficient to determine a realized directional solution response.	State as an exact insufficiency result under declared norms and probes.	A small guard is generally unsafe or harmless.
C02	T1_THEOREM	For common coordinates and right-hand side, the finite response satisfies the exact guarded-inverse identity.	Use proves or satisfies under the assumptions.	Changed residual roots or nonlinear trajectories follow the same identity.
C03	T1_THEOREM	The realized response is bounded by the exact guarded-inverse amplification in a compatible subordinate norm.	Use bounds under the declared norm.	ρ_g is the observed response.
C04	T1_THEOREM	When ρ_1 is below one, the finite response admits the Neumann-series bound with its denominator retained.	Use finite bound with assumptions.	Drop the denominator outside the asymptotic regime.
C05	T1_THEOREM	$\kappa(J)\mu_J$ is a worst-case normwise comparison and not a directional mechanism classifier.	Use as a loose worst-case comparison.	Condition number alone identifies the observed mechanism.
C06	T1_THEOREM	Matched matrix amplitude and worst-direction metrics can yield different realized responses for one fixed probe.	Use proves non-determination.	Typicality, frequency, mesh law, safety, or harm.
C07	T2_VALIDATED_REGISTERED	The registered P2 calculation reproduces the exact metric match and response distinction for six frozen perturbation magnitudes.	Use validates or reproduces.	General operator-family frequency.
C08	T1_THEOREM	A physical-coordinate identity guard generally transforms to $\epsilon D_r^{(-1)} D_x$ rather than the same scalar identity.	Use proves coordinate dependence of the representation.	Every physical identity guard is harmful.
C09	T1_THEOREM	A dimensionless identity guard maps to a generally field-dependent physical perturbation.	Use proves the reverse mapping.	The selected scaling is optimal or uniquely physical.
C10	T2_VALIDATED_REGISTERED	Consistent transformation preserves the registered physical operator and response across the tested coordinate descriptions.	Use validates equivalent transformation in the tested audit.	Universal invariance of independently redeclared guards.
C11	T2_VALIDATED_REGISTERED	Redeclaring the same numeric identity after scaling defines a different physical perturbation for every registered scale factor above one.	Use demonstrates within the frozen audit.	One coordinate system is safer or optimal.
C12	T1_THEOREM	Direct subtraction supports an observability statement only when parent-solution and subtraction-error bounds cover the declared truth lane.	Use conditional on the registered error model.	Direct subtraction is universally unreliable or always reliable.
C13	T2_VALIDATED_REGISTERED	Source-truth and binary-operand truth require separate, non-mixed uncertainty lanes in the registered P4-r1 audit.	Use defines the validated two-truth procedure.	One truth lane can be substituted for the other.
C14	T2_VALIDATED_REGISTERED	The historical direct uncertainty model is reproduced exactly at 287/300 coverage with three severe undercoverage cases.	Use reproduces and discloses.	The guard caused the undercoverage.
C15	T2_VALIDATED_REGISTERED	The revised source and binary envelopes cover all 300 registered development cases with no severe, component, subtraction, or nonfinite failures.	Use validates coverage on the registered development set.	Universal error bound or classification sharpness.
C16	T2_VALIDATED_REGISTERED	The revised source and binary envelopes cover all 300 disjoint registered holdout cases with no severe, component, subtraction, or nonfinite failures.	Use validates coverage on the registered holdout.	Universal coverage or WP27 class validation.
C17	T4_LIMITATION	Coverage validation does not qualify bound tightness or observability-class sharpness.	Use explicitly as a limitation.	Tight, efficient, sharp, or calibrated class thresholds.
C18	T3_BOUNDED_DIAGNOSTIC	A response solve can reduce subtraction error in selected small-guard cases, but it is not an authoritative truth route without its own validated uncertainty bound.	Use motivates an independent response lane.	Stable lane is exact truth or always better.
C19	T3_BOUNDED_DIAGNOSTIC	M8 and M16 are two frozen representative mixed-field tangents to which the metric framework can be applied.	Use illustrates or motivates.	Mesh-independent law, UEL revalidation, or engineering qualification.

ID	Tier	Claim	Allowed strength	Prohibited extrapolation
C20	T3_BOUNDED_DIAGNOSTIC	WP26B reproduced exact capture reconstruction but its selected-iterate subtraction identity failed the registered gate despite machine-level solve residuals.	Use motivates uncertainty-aware observability analysis.	WP26B proved catastrophic cancellation or a guard defect.
C21	T4_LIMITATION	WP27 observability labels and class counts remain diagnostic-only and are excluded from confirmatory Paper 2 conclusions.	Use as an explicit limitation and provenance note.	Use WP27 labels as validated truth or claim counts as classification accuracy.
C22	T4_LIMITATION	Local linear response metrics do not by themselves determine Newton acceptance, iteration count, basin, branch sequence, or final root.	Use does not determine by itself.	Predict nonlinear convergence or basin from η_z , ρ_g , or κ .
C23	T4_LIMITATION	A tangent-only matrix perturbation is not automatically residual, mixed-FE, physical, conservation, or bound-preserving stabilization.	Use numerical guard or matrix perturbation.	Call KSTAB inf-sup stabilization or conservation correction.
C24	T4_LIMITATION	P4-r1 coverage is confined to registered dimension-2/4/8/16 SPD, nonsymmetric, and saddle-point-like decimal families and their disjoint holdout.	Use within the registered families.	Universal uncertainty model or mesh-independent coverage law.
C25	T1_THEOREM_PLUS_T2_VALIDATION	A defensible assessment requires declared coordinates, separate matrix/operator/directional metrics, and uncertainty-qualified response evidence.	Use as the bounded methodological contribution.	Universal guard decision rule, optimal guard, nonlinear prediction, or engineering qualification.
C26	T1_THEOREM	A nonnegative computed response is uncertainty-qualified distinguishable from zero when $d_{\text{obs}} - U > 0$ under a declared truth contract and route-specific uncertainty model.	Use distinguishable from zero only with the truth contract, norm, and uncertainty model declared.	Observable, accurate, important, safe, harmful, or nonlinear consequence without separate evidence.
C27	T2_VALIDATED_REGISTERED	For the frozen M8 binary-operand system, the direct-difference response is distinguishable from zero under its registered direct-route uncertainty envelope.	Use distinguishable from zero and conservative relative-error upper bound for this registered route.	Direct subtraction is generally unreliable, guard safety or harm, physical importance, nonlinear cause, M16 transfer, or mesh law.
C28	T2_VALIDATED_REGISTERED	For the frozen M8 binary-operand system, the response-equation result is distinguishable from zero under its registered response-route uncertainty envelope.	Use distinguishable from zero under this registered response-route envelope.	Universal superiority, guard safety or harm, physical importance, nonlinear cause, M16 transfer, or mesh law.
C29	T2_VALIDATED_REGISTERED	For this frozen M8 system, the response-equation route produced a substantially tighter registered uncertainty envelope than the direct-difference route.	Use tighter for this M8 case only.	Response equation is universally superior, optimal, cheaper, or more reliable.
C30	T4_LIMITATION	The M8 result alone does not transfer to M16; the M16 decision is supported only by its separate registered qualification, and neither case transfers to nonlinear trajectories, mesh behavior, finite-element stability, guard safety or harm, or engineering qualification.	Use independently qualified, repeated on two frozen operators, and case-specific.	Mesh law, dimension trend, population generality, nonlinear classification, safety or harm, or engineering qualification.
C_M16_01	T2_VALIDATED_REGISTERED	The route-specific assessment procedure was completed on a second independently frozen coupled finite-element tangent.	Use completed on a second independently frozen tangent.	Generalized to finite-element systems, mesh trend, dimension law, or engineering qualification.
C_M16_02	T2_VALIDATED_REGISTERED	The M16 direct-difference response was distinguishable from zero under its registered direct-route uncertainty envelope.	Use distinguishable from zero under this registered route envelope.	Direct-route reliability in general, safety or harm, nonlinear consequence, or mesh law.
C_M16_03	T2_VALIDATED_REGISTERED	The M16 response-equation response was distinguishable from zero under its registered response-route uncertainty envelope.	Use distinguishable from zero under this registered route envelope.	Universal route superiority, safety or harm, nonlinear consequence, or mesh law.
C_TWO_OPERATOR_01	T2_VALIDATED_REGISTERED	The same declared assessment architecture was completed for two independently frozen mixed-field tangents.	Maximum wording: repeated on two frozen operators.	Generalization, mesh convergence, dimension dependence, route ranking, safety or harm, nonlinear consequence, or engineering qualification.

Data and code availability

The non-confidential reproducibility artifact is archived in Zenodo at <https://doi.org/10.5281/zenodo.21833851>. The public source repository is <https://github.com/yqma7980/scale-explicit-matrix-perturbation-observability>. The release excludes Abaqus executables, user-element source, solver state histories, and engineering project data.